

EDR vs XDR Research and Comparison

1 Introduction

Modern cybersecurity has shifted from simple prevention to active detection and response. This research analyzes the technical transition from Endpoint Detection and Response (EDR) to the more comprehensive Extended Detection and Response (XDR) systems, evaluating their effectiveness in protecting critical infrastructure.

2 Core Definitions and Evolution

2.1 Endpoint Detection and Response (EDR)

EDR is a security solution that monitors and records activities on endpoints (workstations, servers, mobiles) to detect and respond to advanced threats that bypass traditional antivirus. It focuses on the host level, providing deep visibility into OS processes and local files.

2.2 Extended Detection and Response (XDR)

XDR is the technical evolution of EDR. It is a cross-layered security platform that integrates data from endpoints, networks, cloud environments, and email systems.

2.3 The Evolution: From EDR to XDR

The transition from EDR to XDR marks a move from **Siloed Security** (where data is isolated) to **Unified Security**. While EDR can only see what happens on the device, XDR correlates that data with network traffic and cloud logs to identify complex attack chains like Lateral Movement.

3 Comparison Criteria

To evaluate these systems, we use three primary technical pillars:

1. **Detection Scope:** EDR is limited to host-level telemetry, whereas XDR covers the entire digital ecosystem (Network, Cloud, Identity).

2. **Detection Methods:** EDR relies on behavioral analysis of individual hosts. XDR utilizes AI-driven correlation to link multiple suspicious events across different layers.
3. **Response Capabilities:** EDR response is localized (e.g., isolating a laptop). XDR enables coordinated responses, such as blocking a malicious IP at the firewall while simultaneously resetting an compromised user account.

4 Security Solutions Overview

4.1 EDR Solutions

- **Open-Source Solutions:**
 - **Wazuh:** A powerful open-source platform for threat detection and compliance monitoring, widely used in research labs.
 - **OpenEDR:** Provides comprehensive endpoint telemetry and visibility.
- **Commercial Solutions:**
 - **CrowdStrike Falcon:** A leader in cloud-native endpoint protection.
 - **SentinelOne:** Known for its automated AI-driven response.

4.2 XDR Solutions

- **Commercial Solutions:**
 - **Palo Alto Cortex XDR:** The industry standard for integrating network, cloud, and endpoint data.
 - **Trend Micro Vision One:** Offers advanced cross-layer correlation and threat hunting.

5 Analysis and Recommendations

5.1 When to Choose EDR vs. XDR

- **Choose EDR:** If the organization is small, has a limited budget, and primarily needs to protect remote workers' devices.
- **Choose XDR:** Mandatory for high-security sectors (e.g., Ministry of Defense) where APTs (Advanced Persistent Threats) are a risk and correlation across the network is vital.

5.2 Market Trends and Selection Factors

The market is rapidly moving towards XDR as organizations seek to reduce "Alert Fatigue" by using AI to filter out noise and focus on correlated high-priority incidents.

6 References

1. Gartner Market Guide for Extended Detection and Response.
2. NIST Special Publication 800-61 Revision 2: Computer Security Incident Handling Guide.
3. MITRE ATT&CK Framework for Endpoint and Network Analysis.